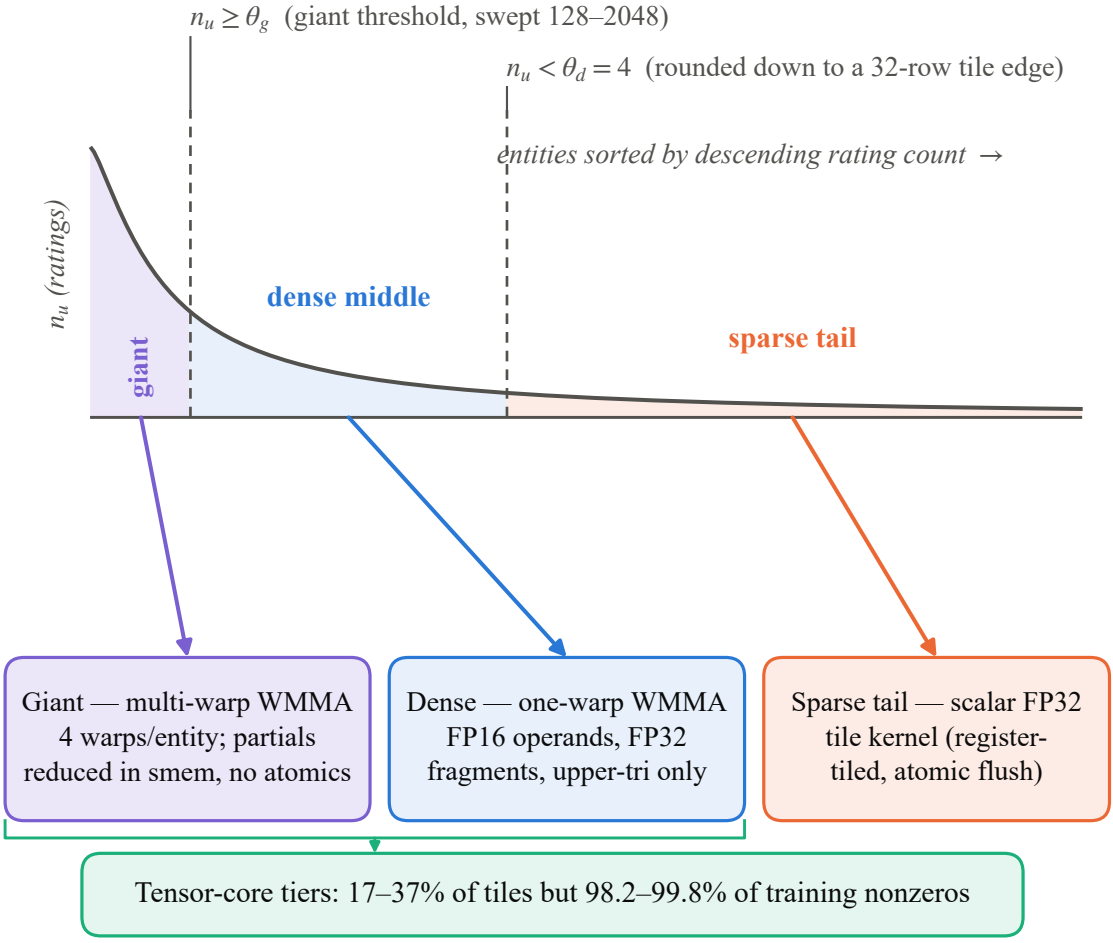

$n_u \geq \theta_g$ (giant threshold, swept 128–2048)
$n_u < \theta_d = 4$ (rounded down to a 32-row tile edge)
entities sorted by descending rating count →
n_u (ratings)
giant
dense middle
sparse tail
Giant — multi-warp WMMA 4 warps/entity; partials reduced in smem, no atomics
Dense — one-warp WMMA FP16 operands, FP32 fragments, upper-tri only
Sparse tail — scalar FP32 tile kernel (register-tiled, atomic flush)
Tensor-core tiers: 17–37% of tiles but 98.2–99.8% of training nonzeros